

Short Papers

Projection and Geodesic-Based Pipe Routing Algorithm

Chengen Wang and Qiang Liu

Abstract—Pipe assembly that connects over hundreds of various pipes to mechanical structures is a fundamental challenge to complex product developments. This paper presents a pipe routing algorithm for automatic generation of pipe assembly plans within a constrained space between inner and outer casings of an aeroengine. In this paper, pipes are first ordered in accordance with interference degrees and other engineering rules. Then, a projection-based routing heuristic is developed to find satisfactory collision-free pipe paths in 2D or 3D spaces. Further, the heuristic is modified, by replacing projections with geodesics, to meet requirements of routing pipes on surfaces. Numerical computations of a 119 pipe routing problem are performed to demonstrate the effectiveness and efficiency of the proposed pipe routing algorithm.

Index Terms—Aeroengine, geodesics, pipe routing, projection, sequence planning.

I. INTRODUCTION

Assembly extensively draws attention from both academic researchers and practitioners in engineering disciplines. While most research efforts have been put on planning assembly sequences and assembly paths for rigid mechanical components; pipe assembly planning has emerged as a new cutting-edge frontier. Unlike conventional assembly planning problems, pipe assembly has to tackle many ubiquitous challenges, such as irregular obstacles, shape variations of components, and tightly constrained spaces. These unique characteristics make it much difficult (if not impossible) to directly implement planning algorithms commonly used in mechanical structure assembly to solve pipe assembly planning problems.

Pipe route planning that looks for collision-free pipe routes in a constrained space is of essential significance to complex product developments. For instance, several hundreds of pipes and cables have to be appropriately routed in an aeroengine, which exerts considerable impacts on product performance, reliability and safety. To address the engineering requirements, many commercial CAD systems commence to provide pipe routing modules allowing engineers efficiently develop and illustrate pipe layouts on screens. However, pipe routing problems are so complicated that nowadays CAD systems can merely support graphical representations of plans conceived on basis of human experience and know-how. Cost-effective pipe routing algorithms that enable automatic generation of pipe layouts still remain an open area of research.

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During the past decades, some pipe routing algorithms have been developed for automatic generation of pipe routes [1]–[12], which are classified, by Park [7], as cell-decomposition methods (e.g., [1], [6], and [10]) and optimization network techniques (e.g., [3], [4]). According to [7], main disadvantages of cell-decomposition methods are exponential explosions of computations with the increase of cell numbers and their incapability of adapting to variable pipe diameters. Typically, the maze algorithm [1] requires a lot of memory to store and process cell data, and becomes considerably time-consuming in 3D spaces. On the other hand, the critical setback of optimization network techniques is that path nodes have to be determined in advance. In this vein, the visibility graph [13] needs to define numerous nodes in a space and introduce lots of additional vertices along edges of obstacles in a 3D space.

Moreover, many algorithms make assumption about obstacle shapes, as polygons or polyhedra [13], or have to process planning spaces before the algorithms run, e.g., constructing configuration-spaces [17]–[21].

Path planning is addressed in many fields, such as plant layout [3], aerospace facilities [5], ship-building [7], [10], robot path planning [13], [14], wire-harnessing in integrated circuits [15], [16], and product disassembly planning [17]–[19].

In this paper, pipe routing is studied in the context of an aeroengine development, where over 200 pipes need to be assembled within a rotational space with irregular obstacles. The study aims at developing a pipe routing algorithm that automatically generates optimal pipe routing plans with respect to shortest pipe lengths, while meeting a number of engineering rules of assembly easiness, stability, and maintenance.

The remainder of this paper is arranged as follows. Section II presents a pipe routing sequence planning method, which transforms a multiple pipe routing problem into a set of serialized single pipe routing problems. Section III presents the projection-based pipe routing algorithm that generates pipe paths between two given points in a constrained space. In Section IV, the algorithm is extended, by introducing approximate geodesic lines, to adapt to requirements of routing pipes on surfaces of revolution. In Section V, numerical computations are performed on a 119 pipe routing problem. Section VI finally concludes this paper.

II. ROUTING SEQUENCE PLANNING

Prior to pipe route planning, the sequence in which pipes are selected needs to be determined. As lately being routed pipes are strongly constrained by previously routed pipes, pipe routing sequences exert remarkable impacts on final pipe layouts. However, it is very difficult to formulate and solve the pipe routing sequence planning problems with analytical mathematic methods. In this paper, pipe routing sequences are generated in alignment with heuristic rules extracted from engineering practices.

First, we subdivide the pipe assembly space between an aeroengine's inner and outer casings into a number of layered subspaces, enclosed within rotational surfaces generated by different meridians. The first pipe routing rule is that pipes should be progressively paved from inner layers to outer layers, with an objective to minimize the number of pipes crossing layers.

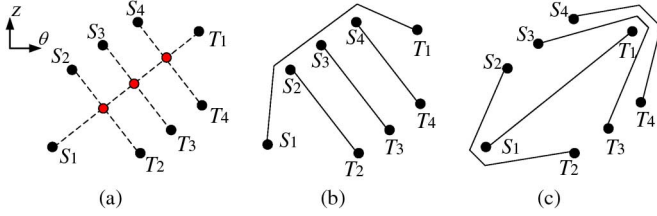


Fig. 1. Interference-based pipe routing sequences.

Pipe routings are fundamentally challenged by intensive interferences among pipe paths, this naturally leads to developing a rule that measures the degree a pipe connection may interfere with other pipe connections. Given a pipe routing problem, we first connect each pair of a start point and a target point with a line; and, a pipe's degree of interference is measured by the number of its connection line intersects with other connection lines.

For instance, as shown in Fig. 1(a), the pipe $\overline{S_1T_1}$'s interference degree is 3, and all other interference degrees are 1. It is observed that the pipe configuration obtained by first routing pipes with smaller interference degrees, as shown in Fig. 1(b), is more satisfactory than that obtained by first routing pipes with larger interference degrees, as shown in Fig. 1(c). Consequently, this observation gives a rise to the second rule governing pipe routing sequence: pipes with smaller interference degrees be first routed.

Similarly, other engineering know-hows in pipe assembly lead to the following pipe routing sequence planning rules.

- Thick pipes need to be routed before the thin ones, in order to reduce the lengths of thick pipes and provide supports for thin pipes.
- Pipes with short connection distances should be routed before those with longer connection distances.
- Pipes that require frequent inspections and maintenance should be routed on the outer layer, which means that these pipes will be routed last.

In this study, the pipe routing sequence planning rules are flexibly applied in several ways. For instance, the rules may be either applied in an order, or in a weighted linear combination.

III. PROJECTION-BASED PIPE ROUTING HEURISTIC

A. Routing Heuristic Based on Projection

The core of the pipe routing algorithm developed in this paper is a heuristic that seeks satisfactory pipe paths in 3D spaces by using a projection principle.

As shown in Fig. 2, let S and T , respectively, denote the start and target positions. Assuming an obstacle (the polyhedron $ABCDE$) exists on the line between the two points, then, an obstacle projection is obtained on the plane through the point T and perpendicular to line $\overline{ST}$.

Further, a point H on the outline of the projected polygon $B_1C_1D_1E_1$ is selected, which has the shortest distance to the target point. The intersection point M of the line $\overline{SH}$ and the obstacle is called **minimum projection angle point**. Obviously, the path $S \rightarrow M \rightarrow T$ yields the shortest distance among all collision-free paths.

Consequently, the basic principle of the routing heuristic is to find a **minimum projection angle point**, through which constituting a collision free path between the start and target points.

Obviously, this heuristics applies to all kinds of obstacles as the projection process assumes no shape constraints.

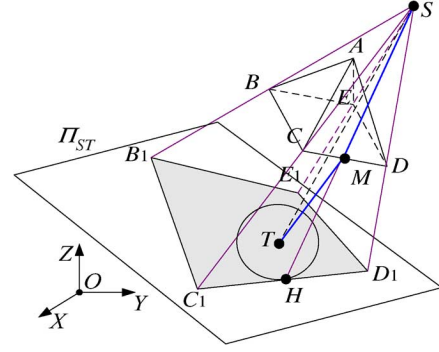


Fig. 2. Projection-based collision-free path.

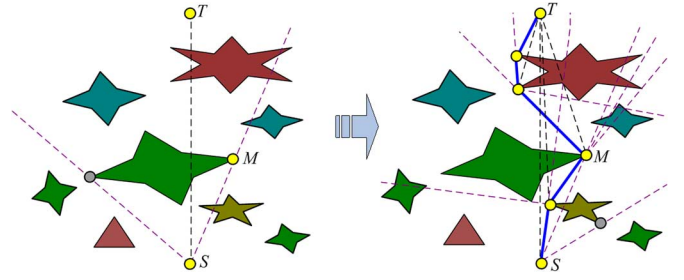


Fig. 3. Pipe routing by individual searching strategy.

B. Pipe Route Searching Strategies

To deal with a pipe engineering problem in a space constrained with numerous obstacles, the pipe routing algorithm applies the projection heuristic with two searching strategies.

The first searching strategy handles only one obstacle in each step, progressively find paths winding through obstacles. The first step begins with winding around the first obstacle on the line between the start point and the target point, using the minimum projection angle heuristic. In each step, a projection line is allowed to penetrate obstacles other than the obstacle of interest. If projection line from the start point to the **minimum projection angle point** found on the obstacle does not penetrate any obstacle, the **minimum projection angle point** will be used as the start point of next step, and the searching strategy repeatedly applies the projection heuristic in the following steps until the final target is reached.

However, in case one or more obstacles obtrude the projection line from the start point and the newly found **minimum projection angle point** in a step, the **minimum projection angle point** will be used as an intermediary target of the next step, and the projection heuristic will be applied to find a path bypassing the emerged obstacles to the intermediary target. After finding a collision-free path to the intermediary target point, the searching strategy uses the intermediary target as the next start point. As shown in Fig. 3, this strategy progressively explores a path from the start point to the final target, bypassing one obstacle in each step.

Alternatively, the second strategy collectively handles obstacles, not allowing projection to penetrate obstacles. As shown in Fig. 4, the searching begins with drawing a projection line between the start and target points, and makes it scan in both clockwise and anticlockwise directions until no collisions are detected. Thus, one extreme point is found on each side, and the point with minimum projection angle is chosen as the star point of next step. The next step repeatedly searches a path to the target point from the new starting point. Finally, the polygon line linking the start point, the intermediary points and the target point constitutes a feasible pipe path.

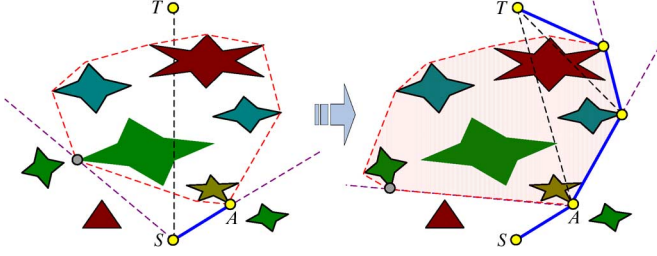


Fig. 4. Pipe routing by collective searching strategy.

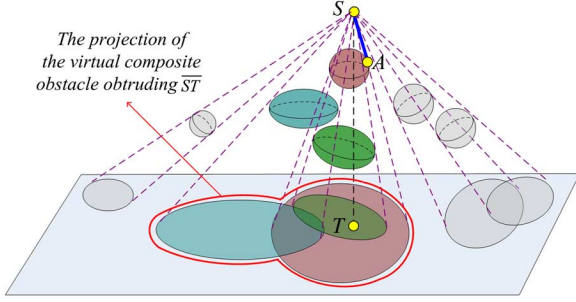


Fig. 5. Pipe routing by collective searching strategy in a 3D space.

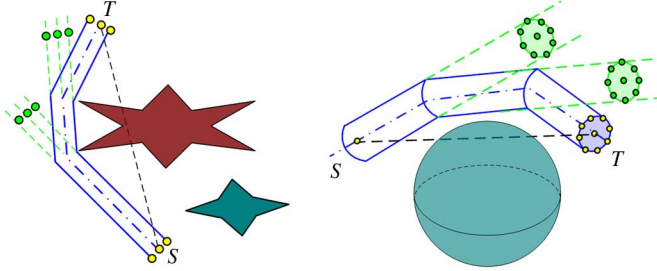


Fig. 6. Consideration of pipe thickness. (a) In a 2D space. (b) In a 3D space.

In respect to 3D cases, as shown in Fig. 5, obstacles are normally projected to the plane through target point, and obstacles with overlapped projections are merged as virtual composite obstacles. Subsequently, a *minimum projection angle point* A can be found on the virtual composite obstacle obtruding the projection from the start point to the target point. The *minimum projection angle point* is used as the next start point, and normal projections of obstacles are made from the new start point, in order to find the next *minimum projection angle point*. The procedure is repeated until the target point is reached.

More practically, to consider the constraint of pipe thickness, we use three parallel projection lines or projection lines of a group of points evenly distributed on a circle for routing a pipe in a 2D or 3D space, as shown in Fig. 6.

IV. GEODESIC-BASED PIPE ROUTING HEURISTIC

Many pipe assembly problems require pipe routes be configured within surfaces, more typically surfaces of revolution, which make pipes be installed closer to mechanical components. One remarkable advantage of surface pipe routing is that the pipe configurations thus generated outperform other pipe configurations in terms of stability and reliability.

However, in cases of surface pipe routing, the concept of normal projection mentioned above becomes inappropriate. To breakthrough this technical barrier, we replace the projection lines with geodesic lines,

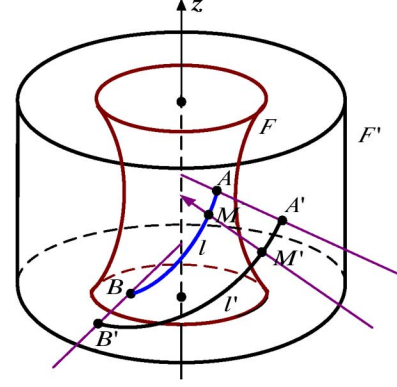


Fig. 7. Formulation of the approximate geodesic.

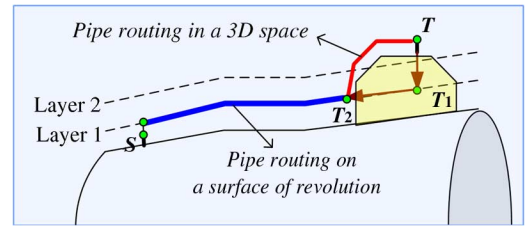


Fig. 8. Hybrid pipe routing case.

which results in the shortest path between two points on a surface. The modified pipe routing heuristic uses geodesic lines, instead of straight lines, to search for pipe routes on a surface of revolution.

Theoretically, geodesics are usually formulated as partial differential equations that require complicated numerical computation methods. To reduce the computation complexity, we develop an *approximate geodesic* model that can be formulated with arithmetic equations and time-efficiently solved.

To this end, the geodesic line in a surface of revolution is approximated by a radial projection of a geodesic in a cylindrical surface. More specifically, as shown in Fig. 7, the geodesic line between points A and B is approximately formulated as the inverse radial projection of the geodesic line connecting points A' and B' on a cylindrical surface with a given radius R' . According to differential geometry [22], the geodesic line l' on the cylindrical surface F' is given as

$$\begin{cases} x_i = R' \cdot \cos(\theta_i) \\ y_i = R' \cdot \sin(\theta_i) \\ z_i = z_{A'} + (z_{B'} - z_{A'}) \cdot (\theta_i - \theta_{A'}) / (\theta_{B'} - \theta_{A'}) \end{cases} \quad (1)$$

where $\theta_i \in [\theta_{A'}, \theta_{B'}]$.

Then, the *approximate geodesic line* l in the surface F determined by meridian curve $\rho = f_C(z)$ is formulated by (2)

$$\begin{cases} z_i = z_A + (z_B - z_A) \cdot (\theta_i - \theta_A) / (\theta_B - \theta_A) \\ x_i = f_C(z_i) \cdot \cos(\theta_i) \\ y_i = f_C(z_i) \cdot \sin(\theta_i) \end{cases} \quad (2)$$

where $\theta_i \in [\theta_A, \theta_B]$.

Apparently, *approximate geodesic lines* on a cylindrical surface, formulated by (2), are geodesic lines, and, approximate geodesic lines connecting points with same longitudes, on a surface of revolution, are identical with the meridians, the real geodesic lines.

Furthermore, meridians of surfaces of revolution are approximately formulated by the least square method over discrete points extracted from aeroengine jacket contours.

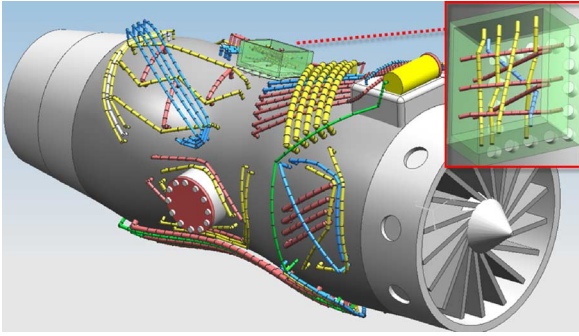


Fig. 9. Pipe routing results.

TABLE I
ALGORITHM COMPUTATION TIME

Routing 119 pipes	on surfaces: 87pipes	in 3D spaces : 22 pipes	in 3D spaces& surfaces: 10 pipes
Time (min)	16.8	9.6	2.3

In general, a pipe can be routed either by applying the normal projection-based heuristic, or the extended geodesic-based heuristic. Nevertheless, there exist more complex cases where one of a pipe's terminals is located on an engine's inner jacket and the other one is on an accessory or the outer casing. In such cases, the projection-based routing heuristic and the geodesic-based routing heuristic have to be applied in conjunction. As shown in Fig. 8, the terminal T on an accessory is radially projected on the inner surface, and a geodesic is drawn to connect the terminal S and the projection point T_1 , which yields an intersection T_2 with the accessory. The geodesic connecting the terminal S and the intermediary point T_2 constitute the first segment of the pipe path. The complementary path connecting the intermediary point T_2 and terminal T is generated by applying the projection-based routing heuristic.

V. NUMERICAL COMPUTATIONS

The *projection and geodesic-based pipe routing algorithm* is implemented on a portable computer (Genuine Intel(R) CPU T1600 @ 1.63 GHz, 1 GB RAM), with customized developments of MATLAB and UG systems.

Numerical computations are performed on a 119 pipe routing problem, within a constrained space formulated in accordance with a simplified aeroengine model. First, the geometric information of the pipe routing space constrained within an aeroengine jacket and outer casing is extracted by using UG/Open GRIP. The extracted pipe routing space is numerically represented as cells, divided by a resolution of $318 \times 50 \times 231$ in the θ , ρ and z axes. Subsequently, information on pipes, connections terminal and other requirements are specified in the context of the routing space. Then, the pipe routing order and pipe paths are computed by applying the routing sequence planning algorithm and the pipe routing algorithm in a MATLAB system. Finally, the planning results are transferred back to the UG system for visualization of pipe layouts, as shown in Fig. 9.

The total planning time is about 29.2 min, about 0.5 min for planning routing sequence, and 28.7 min for pipe routing (see Table I), and a previous study took 5 h to solve a 25 pipe routing problem by using GA approach in [12]. The numerical computation demonstrates that the algorithm presented in this paper is very time efficient, averaging about 15 s for routing one pipe.

TABLE II
PGPRA VERSUS VG

Routing on surfaces	5 pipes	10 pipes	20 pipes
<i>PGPRA</i> (min)	0.5	1.6	3.2
<i>VG</i> (min)	21.6	105.4	279.3

To make a further comparison, different sized pipe routing problems are respectively computed by using visibility graph (*VG*) [13] and the *projection and geodesic-based pipe routing algorithm (PGPRA)* proposed in this study, where the optimal paths in *VG* are found by using Dijkstra's algorithm [23]. As shown in Table II, the PGPRA significantly outperforms the visibility graph method in respect to computation efficiency.

VI. CONCLUSION

A pipe routing algorithm is presented in this paper, which aims at automatically generating satisfactory pipe routing paths in aeroengine assemblies. As an integral part of the routing algorithm, a pipe routing sequence planning method is developed, which takes into account of the intensity of pipe connection interactions as well as engineering rules. The routing sequence planning method provides a well preprocessed partial solution, and allows the pipe routing heuristics to concentrate on seeking collision-free pipe paths.

The *projection and geodesic-based pipe routing algorithm* is able to deal with a variety of pipe routing problems, in 2D and 3D spaces, and on surfaces of revolution. For surface routing problems, introducing *approximate geodesic* significantly reduces computation complexity.

Unlike previous studies where obstacles had to be presented by polygons or polyhedra [13], this study makes no assumptions about obstacles shapes. In addition, the algorithm does not need to process planning spaces in advance, as it did in [13] and [17]–[21], neither need to set up path nodes in advance, as did it in [3], [4], and [13]. These improvements make the algorithm more adaptable to practical engineering scenarios. The proposed routing algorithm also exhibits flexibility in handling pipes with large diameter variations, which considerably challenged previous pipe routing methods.

Finally, even though the algorithm presented in this paper cannot guarantee shortest pipe lengths, it significantly outperforms previous algorithms in terms of problem sizes and computation efficiency, finding solutions with satisfactory pipe lengths as illustrated by the numerical computations.

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Operations Sequencing in Flexible Production Lines With Bernoulli Machines

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Abstract—In flexible manufacturing environments, the performance of a production system is often affected by the sequence of operations. While performance evaluation, improvement and lean design of production systems have been studied extensively, the joint effect of productivity and quality parameters on operations sequencing remains practically unexplored. Indeed, determining the optimal operation sequence has significant implications from both practical and theoretical perspectives. In this paper, in the framework of Bernoulli reliability and quality models, we develop effective indicators that are simple and easy to implement in practice to determine the optimal operation sequence that maximizes the system's production rate.

Note to Practitioners—This paper shows how one can determine the optimal sequence in a two-machine-one-buffer serial line based on indicators, which are simple functions of the productivity and quality parameters of the machines and the capacity of the buffer. It demonstrates that the machine with higher efficiency and/or lower quality should be placed upstream to maximize the throughput. Finally, it shows that if the buffer is sufficiently large, then the machine with higher efficiency should always be placed upstream regardless of the quality performance of the machines.

Index Terms—Bernoulli machine, operations sequencing, production lines, throughput.

I. INTRODUCTION

Performance evaluation, continuous improvement, lean design, and bottleneck identification of production systems have been studied extensively during the past 50 years (see, for instance, monographs [1]–[9]). In these studies, it is often assumed that all parts are processed according a predetermined sequence of operations. The effect of the operation sequence (OS) on the performance of production systems, however, has not been systematically investigated. Indeed, in flexible manufacturing environments, one type of final product can be produced by several different sequences of operations. Thus, determining the optimal sequence of operations to be performed by all parts is of importance for the entire production process. Practical examples of operations sequencing can be widely found in product assembly [10], as well as in machining processes [11].

The existing literature on this subject can be classified into three groups. The first one is concerned with computer-aided process planning (CAPP). In this group, algorithms based on integer programming [11], simulated annealing [12], genetic algorithm [13], tabu search [14], and particle swarm optimization [15] have been developed to determine the best sequence of operations that optimizes certain objectives (e.g., the total production time, total cost, etc.).

The second group mainly studies the design of multistage supply chains. Specifically, Schraner and Hausman [16] develop an indicator to identify opportunities of operations reversal to minimize the total cost. Papers [10] and [17] analyze operations reversal in two-stage

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